Relations, Equivalence Relations, and Partitions

[See also the beginning of Chapter 7, pp. 140–141, of the text.]

Relations

- A **relation from a set S to a set T** is a subset of $S \times T$.
- A **relation on a set S** is a relation from S to S, i.e., a subset of $S \times S$.
- **Notation:** Given a relation R on S (i.e., a subset $R \subseteq S \times S$), we write "$x \sim y$" if $(x, y) \in R$. A relation can be described by specifying the meaning of "$x \sim y$".
- **Example: Congruence relation modulo m.** Given a positive integer m, the congruence relation modulo m is the relation on the set $S = \mathbb{Z}$ defined by $\boxed{x \sim y \Longleftrightarrow x \equiv y \mod m}$, or equivalently, by the subset $R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \equiv y \mod m\}$ of $\mathbb{Z} \times \mathbb{Z}$.

Equivalence relations and equivalence classes

- **Equivalence relation on a set S:** A relation on S that satisfies the following properties:
  - **Reflexive:** For all $x \in S$, $x \sim x$.
  - **Symmetric:** For all $x, y \in S$, $x \sim y$ implies $y \sim x$.
  - **Transitive:** For all $x, y, z \in S$, $x \sim y$ and $y \sim z$ implies $x \sim z$.
- **Equivalence classes:** If $\sim$ is an equivalence relation on S and $x \in S$, the set
$$E_x = \{y \in S : x \sim y\}$$
is called the **equivalence class of x**. Other notations for the equivalence class E_x of x are the bracket and bar notations, $[x]$ and, $\overline{x}$.
- **Example: Congruence relation modulo 3.** The congruence relation modulo 3 is an equivalence relation with the following equivalence classes:
$$E_0 = \overline{0} = \{0, 3, 6, 9, \dots, -3, -6, -9, \dots\},$$
$$E_1 = \overline{1} = \{1, 4, 7, 10, \dots, -2, -5, -8, \dots\},$$
$$E_2 = \overline{2} = \{2, 5, 8, 11, \dots, -1, -4, -7, \dots\}.$$
Every integer belongs to exactly one of these three classes, and any other equivalence class $\overline{x}$ is equal to one of the above classes. For example, the equivalence class of 3, i.e., $\overline{3}$, is the same as $\overline{0}$.

Partitions of a set

Let S be a set. A collection of (finitely or infinitely many) nonempty subsets $A_1, A_2, \dots$ of S is called a **partition of S** if it has the following properties:

- **No overlaps:** The sets A_i are pairwise disjoint; i.e., $A_i \cap A_j = \emptyset$ if $i \neq j$.
- **Union is all of S:** The union of all the sets A_i in the partition is S, i.e., $A_1 \cup A_2 \cup A_3 \cup \cdots = S$.

Equivalence relations and partitions: The Fundamental Theorem

The following fundamental result connects the seemingly unrelated concepts of an equivalence relation on a set S (which, formally, is a subset of $S \times S$ with certain properties) and a partition on S (which is a collection of subsets of S with certain properties).

- **From equivalence relations to partitions:** Given an equivalence relation on a set S, **the set of distinct equivalence classes forms a partition of S**. That is, we have: (i) For any $x, y \in S$, the associated equivalence classes E_x and E_y are either equal (i.e., $E_x = E_y$) or disjoint (i.e., $E_x \cap E_y = \emptyset$). (ii) For any $x \in S$ there exists an equivalence class that contains x (namely, the class E_x).
- **From partitions to equivalence relations:** Given a partition of S into sets $A_1, A_2, \dots, A_n$, the relation defined by $\boxed{\text{"}x \sim y\text{"} \Longleftrightarrow \text{"}x \text{ and } y \text{ belong to the same set } A_i \text{ from the partition"}}$ is an equivalence relation on S.